

Component List

1. Seeed Studio XIAO ESP32C6

Description: Compact microcontroller with Wi-Fi 6, BLE, and rich GPIOs.

Available Pins:

GPIO0, GPIO1, GPIO2, GPIO16, GPIO17, GPIO18, GPIO19, GPIO20, GPIO21, GPIO22, GPIO23, 3V3, 5V, GND, B+, B-

2. SD Card Module

Description: SPI-based storage module for reading/writing to microSD cards.

Pins: CS, SCK, MOSI, MISO, VCC, GND

3. 0.96" OLED Display (I2C)

Description: Small monochrome OLED screen for visual feedback.

Pins: GND, VDD (3.3V), SDA, SCL

4. Pushbuttons ×3

Description: Tactile pushbuttons for user input.

Pins (per button): Pin 1 (Signal), Pin 2 (GND)

Assigned GPIOs: GPIO0, GPIO1, GPIO2

5. TRRS 3.5mm Jack Breakout

Description: Breakout board for standard 3.5mm stereo (TRRS) audio connector.

Pins: Sleeve (GND), Ring2, Ring1, Tip

6. 1kΩ Resistor

Description: Used for signal conditioning/filtering.

Value: 1000 Ω

Pins: pin1, pin2

7. 0.1 μF Ceramic Capacitor

Description: Used for RC low-pass filtering.

Value: 0.1 μF

Pins: pin0, pin1